Week 4 Day 1 — ML Foundations Summary

1. Problem Framing

Dataset: UCI Adult (Census Income)
Objective: Predict whether an individual earns **more than $50K/year** to support targeted marketing campaigns.
Class Distribution: 76.07% ≤50K | 23.93% >50K
Primary Metric: Precision
Precision was chosen because the business prefers minimizing unnecessary marketing outreach. Correctly identifying high-income customers is more valuable than contacting a large number of unlikely prospects.

2. Data Preparation

The dataset was split using **stratified sampling** into:

- **70% Training**
- **10% Development**
- **20% Hold-Out Test**

A fixed `random_state=42` was used to ensure reproducibility. The hold-out test set was reserved for final evaluation to prevent **test set leakage**.

3. Baseline Results

Model	Accuracy	Precision	Recall	F1
Majority Class	0.761	0.000	0.000	0.000
Rule (`capital-gain > 0`)	**0.782**	**0.625**	**0.225**	**0.331**

The rule-based baseline outperformed the majority-class predictor across all metrics and established the performance benchmark for future machine learning models.

4. Error Analysis

False Positives

- Positive capital gain but lower education or less specialized occupations.
- Capital gain alone was not sufficient to indicate high annual income.

False Negatives

- Zero capital gain despite higher education, professional occupations, and longer working hours.
- The rule failed to identify many genuine high-income individuals.

Next Improvements

- Encode categorical variables.
- Transform skewed numerical features.
- Engineer additional features.
- Group rare categories.
- Train machine learning models that combine multiple features.

5. Conclusion

The **rule-based baseline** achieved the strongest overall performance (**Precision = 0.625, F1 = 0.331**) and will serve as the benchmark for the remainder of the project. Future models should improve **F1-score** while maintaining strong precision to better identify high-income individuals without increasing unnecessary outreach.